

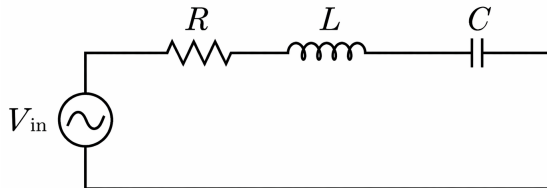
Assignment–6

TRANSIENT AND FREQUENCY RESPONSE OF SERIES RLC CIRCUIT

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Aim

To design and analyze a series RLC circuit and study its frequency and transient response by observing the voltage across the capacitor.



FREQUENCY RESPONSE OF A SERIES RLC CIRCUIT

The response of the circuit depends on the values of R , L , and C .
Applying Kirchhoff's Voltage Law (KVL):

$$V(t) = Ri + L \frac{di}{dt} + \frac{1}{C} \int i dt$$

For natural response:

$$V(t) = 0$$

$$0 = Ri + L \frac{di}{dt} + \frac{1}{C} \int i dt$$

Differentiating with respect to time:

$$0 = R \frac{di}{dt} + L \frac{d^2i}{dt^2} + \frac{i}{C}$$

Rearranging,

$$\frac{d^2i}{dt^2} + \frac{R}{L} \frac{di}{dt} + \frac{1}{LC} i = 0$$

Comparing with the standard second order equation,

$$\frac{d^2x}{dt^2} + 2\zeta\omega_n \frac{dx}{dt} + \omega_n^2 x = 0$$

we obtain

$$2\zeta\omega_n = \frac{R}{L}, \quad \omega_n = \frac{1}{\sqrt{LC}}$$

Substituting ω_n , the damping ratio is

$$\zeta = \frac{R}{2} \sqrt{\frac{C}{L}}$$

Resonant Frequency

At a particular frequency called the **resonant frequency**, the inductive reactance becomes equal to the capacitive reactance.

$$X_L = X_C$$

Since

$$X_L = 2\pi fL, \quad X_C = \frac{1}{2\pi fC}$$

equating them gives

$$2\pi fL = \frac{1}{2\pi fC}$$

which simplifies to the resonant frequency

$$f_r = \frac{1}{2\pi \sqrt{LC}}$$

At resonance, the circuit **impedance is minimum** and the **current is maximum**.

Resonant Frequency Calculation

Given $L = 10 \text{ mH} = 0.01 \text{ H}$ and $C = 470 \text{ pF} = 470 \times 10^{-12} \text{ F}$,

$$f_r = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{0.01 \times 470 \times 10^{-12}}} \approx 7.33 \times 10^4 \text{ Hz}$$

$$f_r \approx 73 \text{ kHz}$$

Bandwidth

The bandwidth of a series RLC circuit is

$$BW = \frac{R}{2\pi L}$$

DAMPING CONDITIONS

Different conditions based on damping ratio are:

- **Underdamped:** $\zeta < 1$
- **Critically damped:** $\zeta = 1$
- **Overdamped:** $\zeta > 1$

1 UNDERDAMPED CONDITION

The respective values are $R = 3.3 \text{ k}\Omega$, $L = 10 \text{ mH}$, and $C = 470 \text{ pF}$.

The damping ratio is

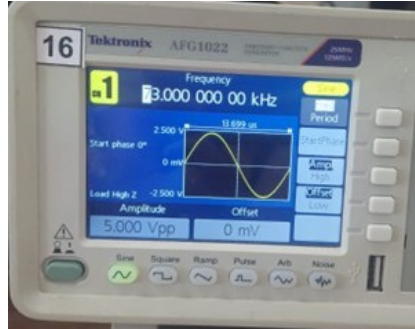
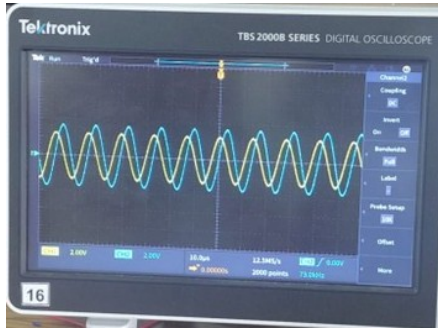
$$\zeta = \frac{R}{2} \sqrt{\frac{C}{L}}$$

$$\zeta = \frac{3300}{2} \sqrt{\frac{470 \times 10^{-12}}{0.01}} \approx 0.358$$

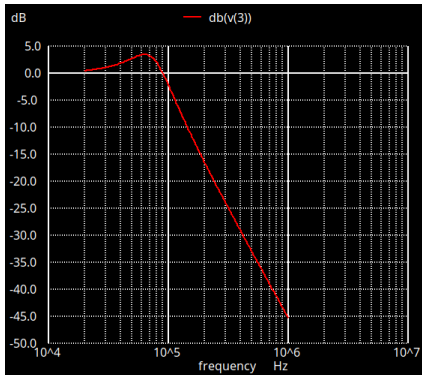
$$\zeta < 1$$

Since $\zeta < 1$, the circuit is **underdamped**.

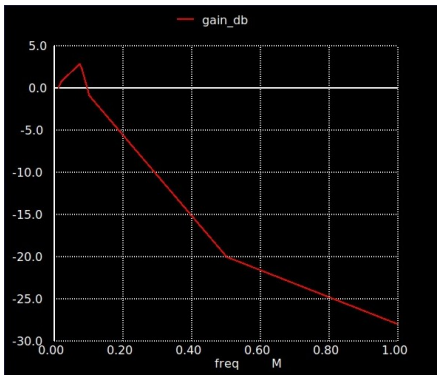
At low frequencies the capacitor reactance is high, giving a larger output voltage. As frequency approaches the resonant frequency, the output voltage increases and reaches a maximum. Beyond resonance, the voltage decreases with increasing frequency, producing a narrow bandwidth and a clear resonance peak.



Plot of Gain vs Frequency



Theoretical



Experimental

The circuit response was experimentally observed at different input frequencies.

- At 25 kHz (below resonance) the output amplitude is moderate.
- At 73 kHz (near resonant frequency) the output voltage reaches its max.
- At 100 kHz the amplitude begins to decrease as the frequency moves beyond resonance.
- At 500 kHz the output voltage becomes significantly smaller due to increased impedance.

2 CRITICALLY DAMPED CONDITION

The circuit parameters are $R = 9\text{ k}\Omega$, $L = 10\text{ mH}$, and $C = 470\text{ pF}$.
The damping ratio is

$$\zeta = \frac{R}{2} \sqrt{\frac{C}{L}}$$
$$\zeta = \frac{9000}{2} \sqrt{\frac{470 \times 10^{-12}}{0.01}} \approx 0.98$$

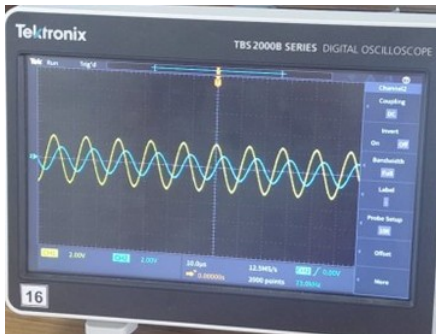
$$\boxed{\zeta \approx 1}$$

Hence, the circuit is **critically damped**.

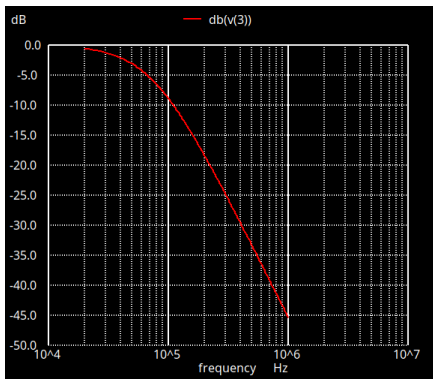
As the frequency increases, the output voltage rises gradually and reaches a moderate value near resonance without a sharp peak. Beyond this region, the output voltage decreases slowly, indicating a stable response with moderate bandwidth.

The circuit response was experimentally observed at different input frequencies.

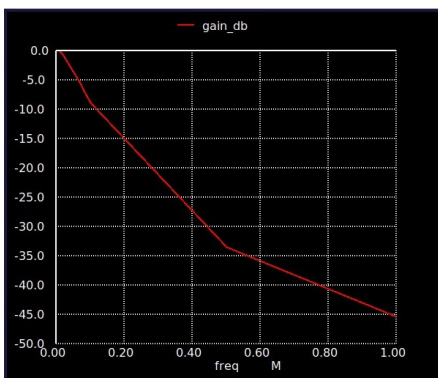
- At 25 kHz (below resonance) the output amplitude is moderate.
- At 73 kHz (near resonant frequency) the output voltage increases but no sharp peak is observed.
- At 100 kHz the amplitude starts to decrease gradually.
- At 500 kHz the output voltage becomes smaller due to increased impedance.



Plot of Gain vs Frequency



Theoretical



Experimental

3 OVERDAMPED CONDITION

The circuit parameters are $R = 18\text{ k}\Omega$, $L = 10\text{ mH}$, and $C = 470\text{ pF}$.

The damping ratio is

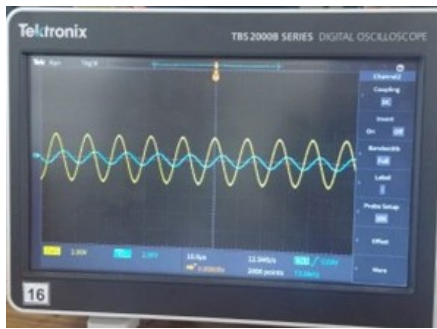
$$\zeta = \frac{R}{2} \sqrt{\frac{C}{L}}$$

$$\zeta = \frac{18000}{2} \sqrt{\frac{470 \times 10^{-12}}{0.01}} \approx 1.96$$

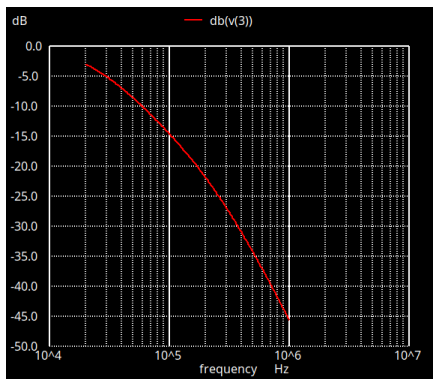
$$\boxed{\zeta > 1}$$

Hence, the circuit is **overdamped**.

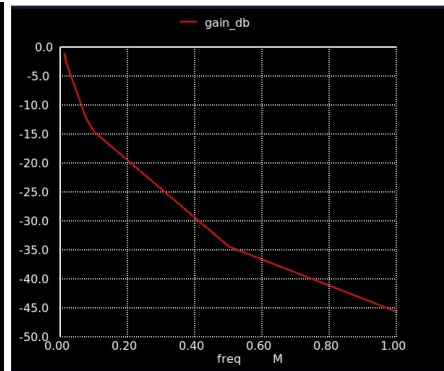
As the frequency increases, the output voltage varies slowly and no sharp peak is observed near the resonant frequency. The frequency response curve appears relatively flat due to the large resistance, which heavily damps the circuit and weakens the resonance effect.



Plot of Gain vs Frequency



Theoretical



Experimental

The circuit response was experimentally observed at different input frequencies.

- At 25 kHz the output amplitude is moderate.
- At 73 kHz the output voltage increases slightly but no resonance peak is observed.
- At 100 kHz the amplitude changes slowly.
- At 500 kHz the output voltage becomes smaller as the impedance increases.

TRANSIENT RESPONSE OF A SERIES RLC CIRCUIT

For the given values:

$$L = 10 \text{ mH}, \quad C = 470 \text{ pF}$$

The natural frequency is given by

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

$$\omega_0 = \frac{1}{\sqrt{(10 \times 10^{-3})(470 \times 10^{-12})}}$$

$$\omega_0 \approx 4.61 \times 10^5 \text{ rad/s}$$

UNDERDAMPED CONDITION

The damped frequency is given by

$$\omega_d = \omega_0 \sqrt{1 - \zeta^2}$$

where

- ω_0 = Natural frequency
- ζ = Damping ratio

Natural frequency:

$$\omega_0 = 4.61 \times 10^5 \text{ rad/s}$$

For underdamped condition:

$$\zeta = 0.358$$

$$\omega_d = 4.61 \times 10^5 \sqrt{1 - (0.358)^2} = 4.61 \times 10^5 \sqrt{0.872}$$

$$\omega_d \approx 4.30 \times 10^5 \text{ rad/s}$$

The time response of an underdamped second order system is given by

$$x(t) = e^{-\zeta\omega_0 t} [A \cos(\omega_d t) + B \sin(\omega_d t)]$$

where

- ζ = Damping ratio
- ω_0 = Natural frequency
- ω_d = Damped frequency
- A, B = Constants determined by initial conditions

Settling Time

Settling time is the time required for the response to reach and remain near the steady-state value.

$$T_s = \frac{4}{\zeta\omega_0}$$

For $R = 3.3k\Omega$, $L = 10mH$, and $C = 470pF$:

$$\zeta = 0.358 < 1 \quad (\text{underdamped})$$

$$T_s = \frac{4}{0.358 \times 4.61 \times 10^5} \approx 2.42 \times 10^{-5} \text{ s} = 24.2 \mu s$$

Peak Time

Peak time is the time taken for the response to reach its first maximum value.

$$T_p = \frac{\pi}{\omega_d}$$

$$T_p = \frac{3.1416}{4.30 \times 10^5} \approx 7.3 \mu s$$

Maximum Overshoot

Maximum overshoot is the amount by which the response exceeds the final steady-state value after a step input.

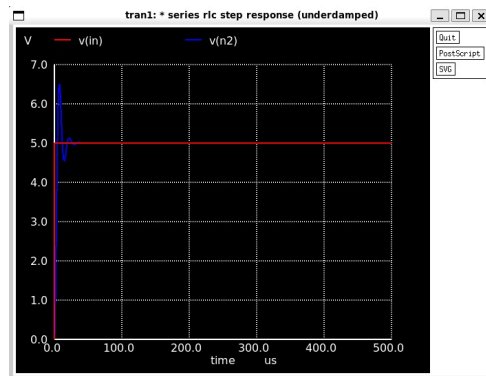
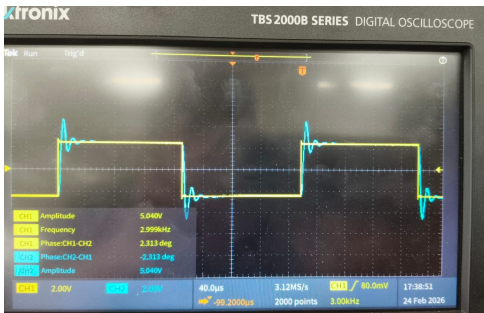
$$M_p = e^{\frac{-\zeta\pi}{\sqrt{1-\zeta^2}}}$$

For $\zeta = 0.358$,

$$M_p = e^{\frac{-0.358\pi}{\sqrt{1-(0.358)^2}}}$$

$$M_p \approx 0.30 \text{ (30\%)}$$

Observation



NGSPICE simulation waveform

The waveform exhibits oscillations before reaching steady state and shows an overshoot above the final value. The oscillations gradually decay due to energy dissipation in the resistor, and the system settles to the steady-state value after a few cycles. This behavior confirms that the circuit is underdamped.

CRITICALLY DAMPED CONDITION

For the given values:

$$R = 9k\Omega, \quad L = 10mH, \quad C = 470pF$$

Critical damping occurs when the resistance is equal to the critical resistance.

$$R_c = 2\sqrt{\frac{L}{C}}$$

$$R_c = 2\sqrt{\frac{10 \times 10^{-3}}{470 \times 10^{-12}}}$$

$$R_c \approx 9.23 \text{ k}\Omega$$

Since the chosen resistance $R = 9 \text{ k}\Omega$ is approximately equal to the critical resistance, the damping ratio is

$$\zeta \approx 1$$

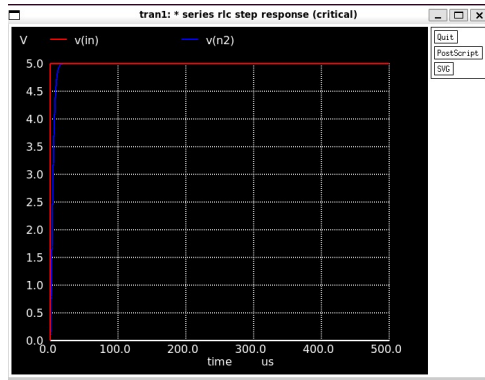
The time response of a critically damped system is given by

$$x(t) = (A + Bt)e^{-\omega_0 t}$$

where A and B are constants determined by the initial conditions.

Observation





NGSPICE simulation waveform

The waveform reaches the steady-state value quickly without oscillations. No overshoot is observed. This represents the critically damped condition, where the system returns to equilibrium in the shortest possible time without oscillating.

OVERDAMPED CONDITION

For the given values:

$$R = 18k\Omega, \quad L = 10mH, \quad C = 470pF$$

Overdamping occurs when the resistance is greater than the critical resistance.

$$R_c \approx 9.23k\Omega$$

Since $R > R_c$, the damping ratio satisfies

$$\zeta > 1$$

Hence the circuit operates in the overdamped region.

The time response of an overdamped second-order system is

$$x(t) = Ae^{s_1 t} + Be^{s_2 t}$$

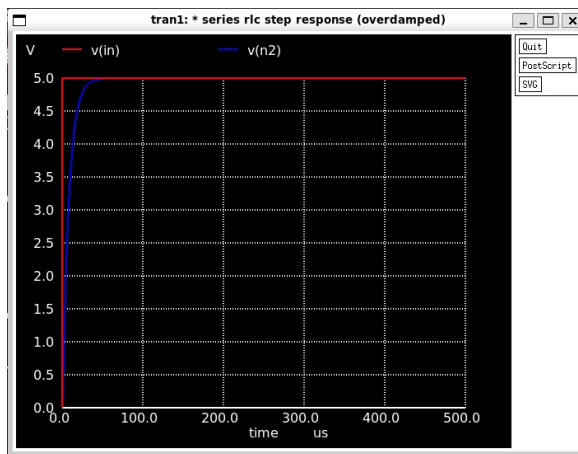
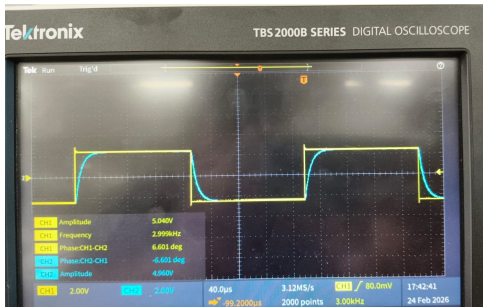
where

$$s_1 = -\omega_0(\zeta + \sqrt{\zeta^2 - 1}), \quad s_2 = -\omega_0(\zeta - \sqrt{\zeta^2 - 1})$$

For $\zeta = 1.95$,

$$s_1 \approx -1.67 \times 10^6, \quad s_2 \approx -1.29 \times 10^5$$

Observation



NGSPICE simulation waveform

The waveform slowly approaches the steady-state value without oscillations. No overshoot is observed. When the resistance is high, the circuit becomes overdamped and energy stored in the reactive elements is dissipated quickly by the resistor, resulting in a slow but stable response.

CORRELATION BETWEEN TRANSIENT AND FREQUENCY RESPONSE

The behavior of an RLC circuit in the time domain is closely related to its behavior in the frequency domain. The damping of the system determines both the transient response and the frequency response characteristics.

- **Underdamped Condition ($\zeta < 1$):** The transient response shows

oscillations and overshoot before settling to the steady state. In the frequency domain, this corresponds to a pronounced resonance peak near the natural frequency.

- **Critically Damped Condition ($\zeta = 1$):** The system reaches the steady-state value quickly without oscillations. In the frequency response, the resonance peak disappears and the response becomes smoother.
- **Overdamped Condition ($\zeta > 1$):** The transient response approaches the steady-state slowly without oscillations or overshoot. In the frequency domain, the resonance effect is absent and the response becomes broad with reduced gain.

Thus, the damping factor controlled by the resistance determines both the transient behavior and the frequency response of the RLC circuit.